

CSE 432 Robotics

Forward Kinematics 1

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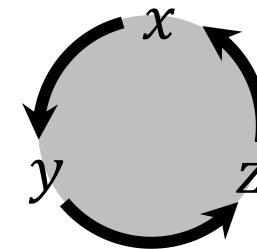
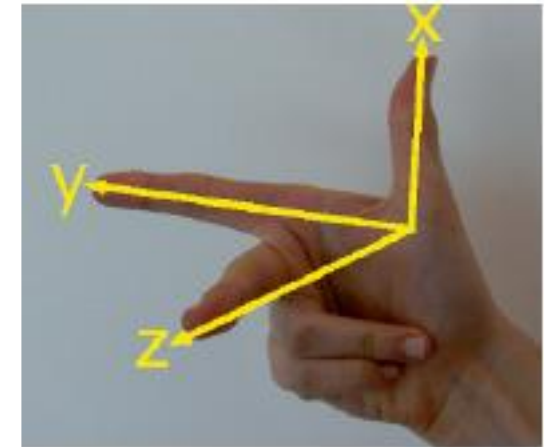
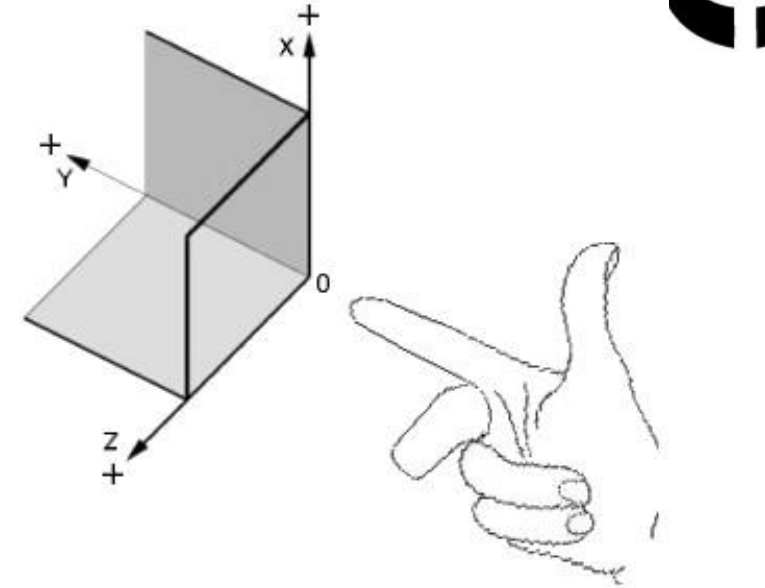
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Kinematics

- The problem of kinematics is to describe the motion of the manipulator without consideration of the forces and torques causing the motion.
- The kinematic description is therefore a geometric one.
- We first consider the problem of forward kinematics, which is to determine **the position and orientation of the end-effector given the values for the joint variables** of the robot.
- In this course, always use right-hand coordinate frame





KINEMATIC CHAINS

- A robot manipulator consists of links connected by joints.
- Joints can be simple (revolute or prismatic) or complex (ball-and-socket).
- Simple joints have one degree of freedom:
 - Revolute: rotation angle
 - Prismatic: linear displacement
- Ball-and-socket joints have two degrees of freedom.
- In this course, all joints are assumed to have a single degree of freedom.



KINEMATIC CHAINS

- This assumption does not involve any real loss of generality, since joints such as a ball and socket joint (two degrees-of-freedom) or a spherical wrist (three degrees-of-freedom) can always be thought of as a succession of single degree-of-freedom joints with links of length zero in between.
- With the assumption that each joint has a single degree-of-freedom, **the action of each joint** can be described by a single real number; the angle of rotation in the case of a revolute joint or the displacement in the case of a prismatic joint.
- The objective of **forward kinematic** analysis is to determine the **cumulative effect of the entire set of joint variables**, that is, to determine the position and orientation of the end effector given the values of these joint variables.



KINEMATIC CHAINS

- A robot manipulator with n joints will have $n + 1$ links, since each joint connects two links.
- We number the joints from 1 to n , and we number the links from 0 to n , starting from the base.
- By this convention, joint i connects link $i - 1$ to link i .
- We will consider the location of joint i to be fixed with respect to link $i - 1$.
- When joint i is actuated, link i moves.
- Link 0 (the first link) is fixed and does not move when the joints are actuated.
- Of course, the robot manipulator could itself be mobile, but we will not consider this case in this course.



KINEMATIC CHAINS

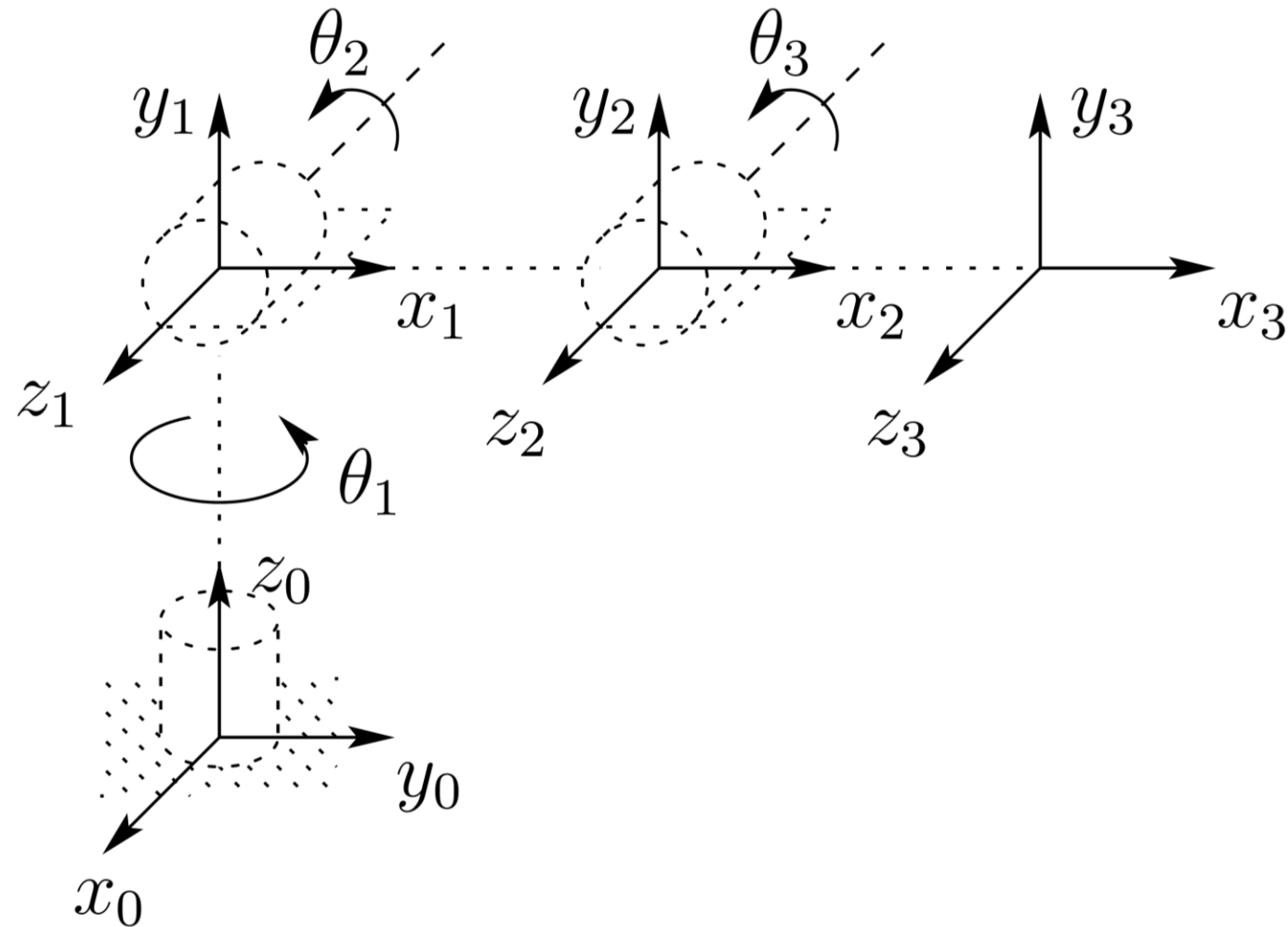
- With the i^{th} joint, we associate a joint variable, denoted by q_i .
- In the case of a revolute joint, q_i is the angle of rotation.
- In the case of a prismatic joint, q_i is the joint displacement.

$$q_i = \begin{cases} \theta_i & \text{if joint } i \text{ is revolute} \\ d_i & \text{if joint } i \text{ is prismatic} \end{cases}$$

- To perform the kinematic analysis, we attach a coordinate frame rigidly to each link. In particular, we attach $o_i x_i y_i z_i$ to link i .
- This means that, whatever motion the robot executes, the **coordinates of each point on link i are constant when expressed in the i^{th} coordinate frame.**
- Furthermore, when joint i is actuated, link i and its attached frame, $o_i x_i y_i z_i$, experience a resulting motion.
- The frame $o_0 x_0 y_0 z_0$, which is attached to the robot base, is referred to as the inertial frame.



KINEMATIC CHAINS





KINEMATIC CHAINS

- Now suppose A_i is the homogeneous transformation matrix that expresses the position and orientation of $o_i x_i y_i z_i$ with respect to $o_{i-1} x_{i-1} y_{i-1} z_{i-1}$.
- The matrix A_i is not constant, but varies as the configuration of the robot is changed.
- However, the assumption that all joints are either revolute or prismatic means that A_i is a function of only a single joint variable, namely q_i .
- In other words

$$A_i = A_i(q_i)$$



KINEMATIC CHAINS

- Now the homogeneous transformation matrix that expresses the position and orientation of $o_j x_j y_j z_j$ with respect to $o_i x_i y_i z_i$ is called, by convention, a transformation matrix.

- Transformation matrix T_j^i can be obtained as follows

$$T_j^i = \begin{cases} A_{i+1} A_{i+2} \dots A_{j-1} A_j & \text{if } i < j \\ I & \text{if } i = j \\ (T_i^j)^{-1} & \text{if } j > i \end{cases}$$

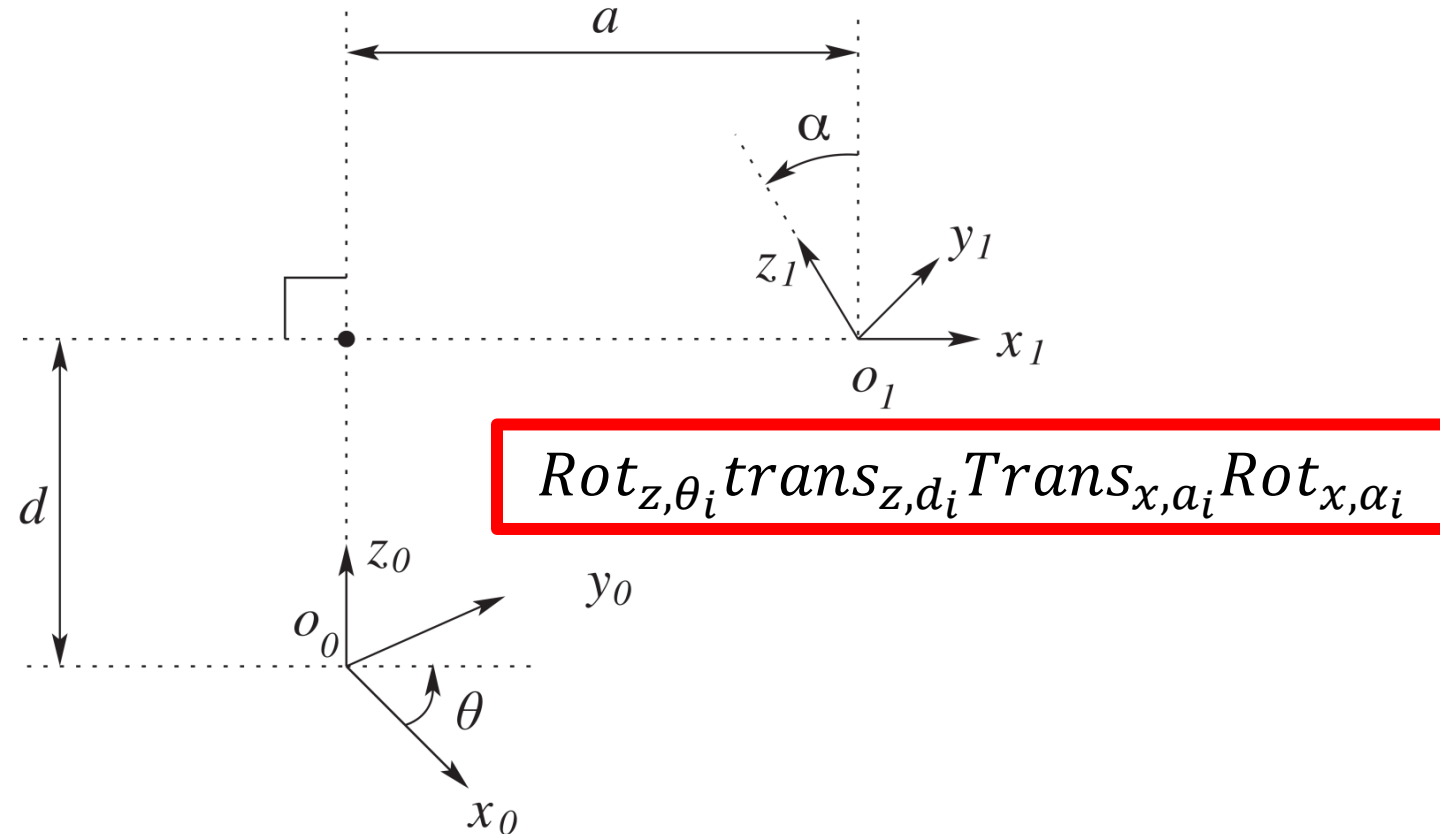
- Each homogeneous transformation A_i is of the form

$$A_i = \begin{bmatrix} R_i^{i-1} & o_i^{i-1} \\ 0 & 1 \end{bmatrix}$$
$$T_j^i = A_{i+1} A_{i+2} \dots A_{j-1} A_j = \begin{bmatrix} R_j^i & o_j^i \\ 0 & 1 \end{bmatrix}$$



FORWARD KINEMATICS: THE DH-CONVENTION

- A commonly used convention for selecting frames of reference in robotic applications is the Denavit-Hartenberg, or DH convention.





FORWARD KINEMATICS: THE DH-CONVENTION

- In this convention, each homogeneous transformation A_i is represented as a product of four basic transformations

$$A_i = Rot_{z,\theta_i} trans_{z,d_i} Trans_{x,a_i} Rot_{x,\alpha_i}$$

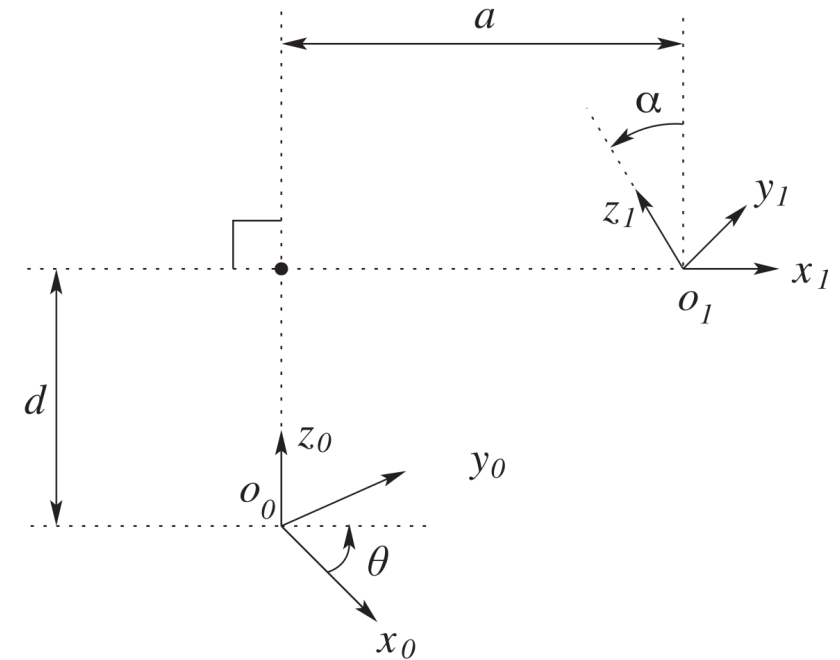
$$A_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



FORWARD KINEMATICS: THE DH-CONVENTION

- θ_i = the angle between x_{i-1} and x_i measured about z_{i-1} .
 - θ_i is **variable** if joint i is **revolute**.
- d_i = distance along z_{i-1} from o_{i-1} to the intersection of the x_i and z_{i-1} axes.
 - d_i is **variable** if joint i is **prismatic**.
- a_i = distance along x_i from the intersection of the x_i and z_{i-1} axes to o_i .
- α_i = the angle between z_{i-1} and z_i measured about x_i .





FORWARD KINEMATICS: THE DH-CONVENTION

- The four quantities θ_i , a_i , α_i and d_i are parameters associated with link i and joint i .

$a_i \rightarrow$ link length

$\alpha_i \rightarrow$ Link twist

$d_i \rightarrow$ link offset

$\theta_i \rightarrow$ joint angle

- Since the matrix A_i is a function of a single variable, it turns out that three of the above four quantities are constant.
- For a given link, while the fourth parameter, θ_i for a revolute joint and d_i for a prismatic joint, is the joint variable.



FORWARD KINEMATICS: THE DH-CONVENTION

- Homogeneous transformation matrix can be characterized by six numbers, such as, for example, three numbers to specify the fourth column of the matrix and three Euler angles to specify the upper left 3×3 rotation matrix.
- In the DH representation, in contrast, there are only four parameters.

How is this possible?

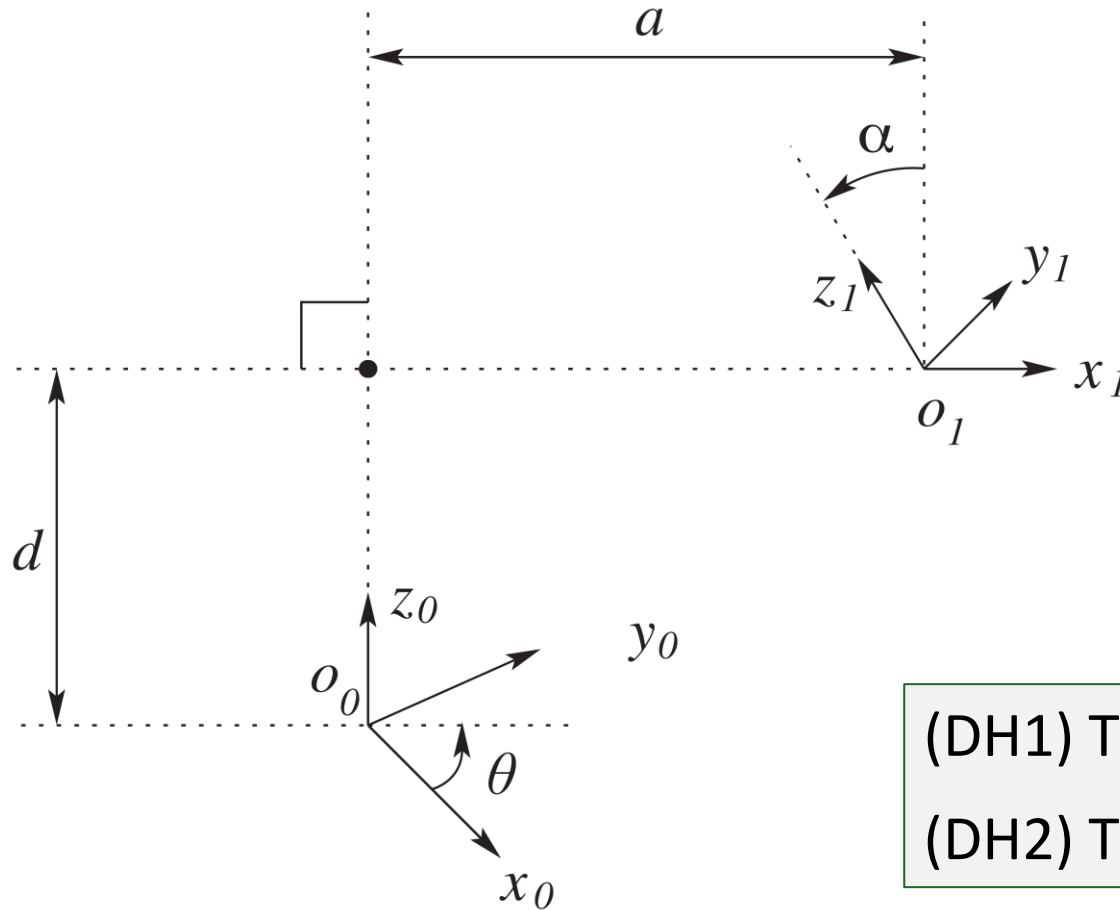


FORWARD KINEMATICS: THE DH-CONVENTION

- The answer is that, while frame i is required to be rigidly attached to link i , we have considerable freedom in choosing the origin and the coordinate axes of the frame.
- For example, it is not necessary that the origin, o_i , of frame i be placed at the physical end of link i .
- In fact, it is not even necessary that frame i be placed within the physical link; frame i could lie in free space — so long as frame i is rigidly attached to link i .
- By a clever choice of the origin and the coordinate axes, it is possible to cut down the number of parameters needed from six to four.



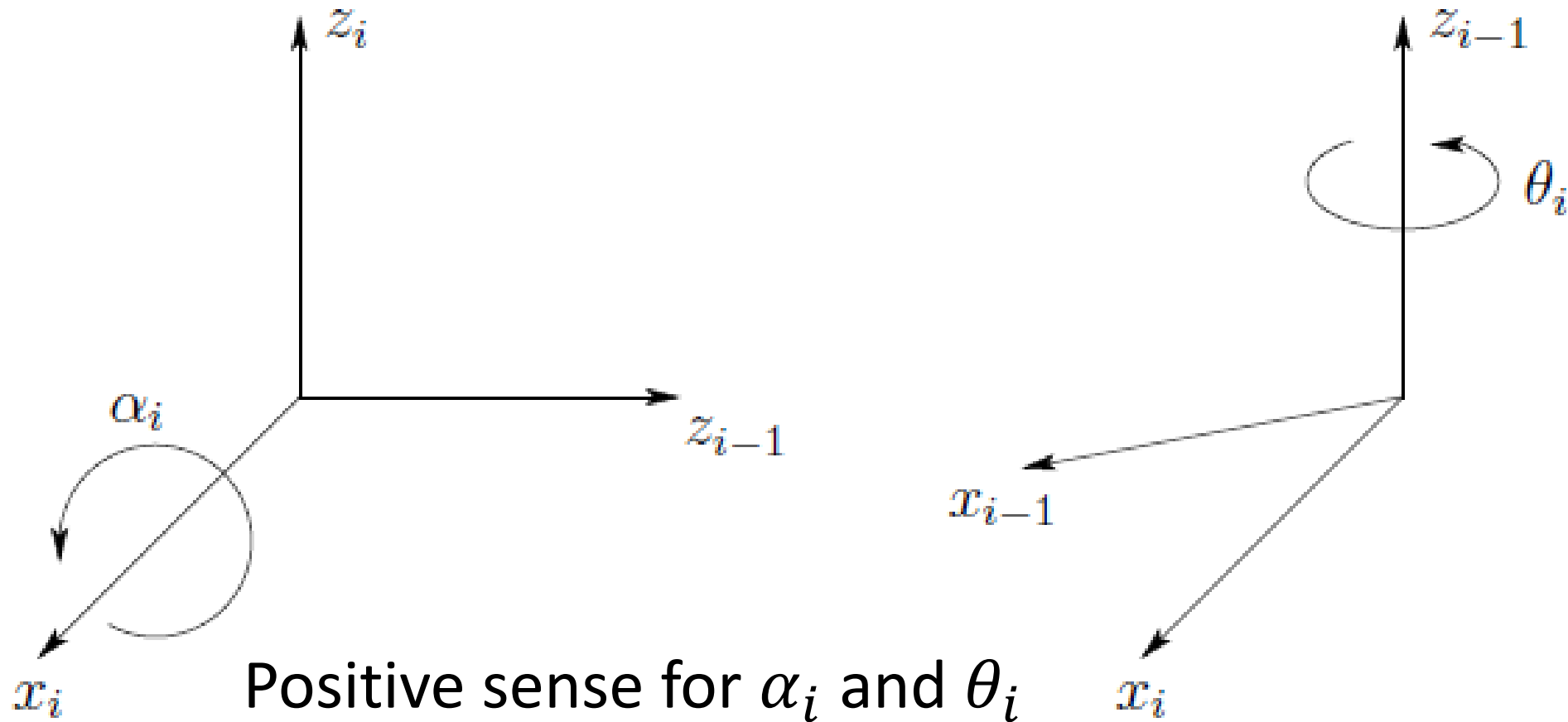
FORWARD KINEMATICS: THE DH-CONVENTION



- (DH1) The axis x_1 is perpendicular to the axis z_0 .
- (DH2) The axis x_1 intersects the axis z_0 .



FORWARD KINEMATICS: THE DH-CONVENTION





Assigning the coordinate frames

- For a given robot manipulator, one can always choose the frames $0, \dots, n$ in such a way that the D-H conditions are satisfied.
- The choices of the various coordinate frames are not unique, even when constrained by the requirements above.
- Thus, it is possible that different engineers will derive differing, but equally correct, coordinate frame assignments for the links of the robot.
- The end result (i.e., the matrix T_n^0) will be the same, regardless of the assignment of intermediate link frames (assuming that the coordinate frames for link n coincide).



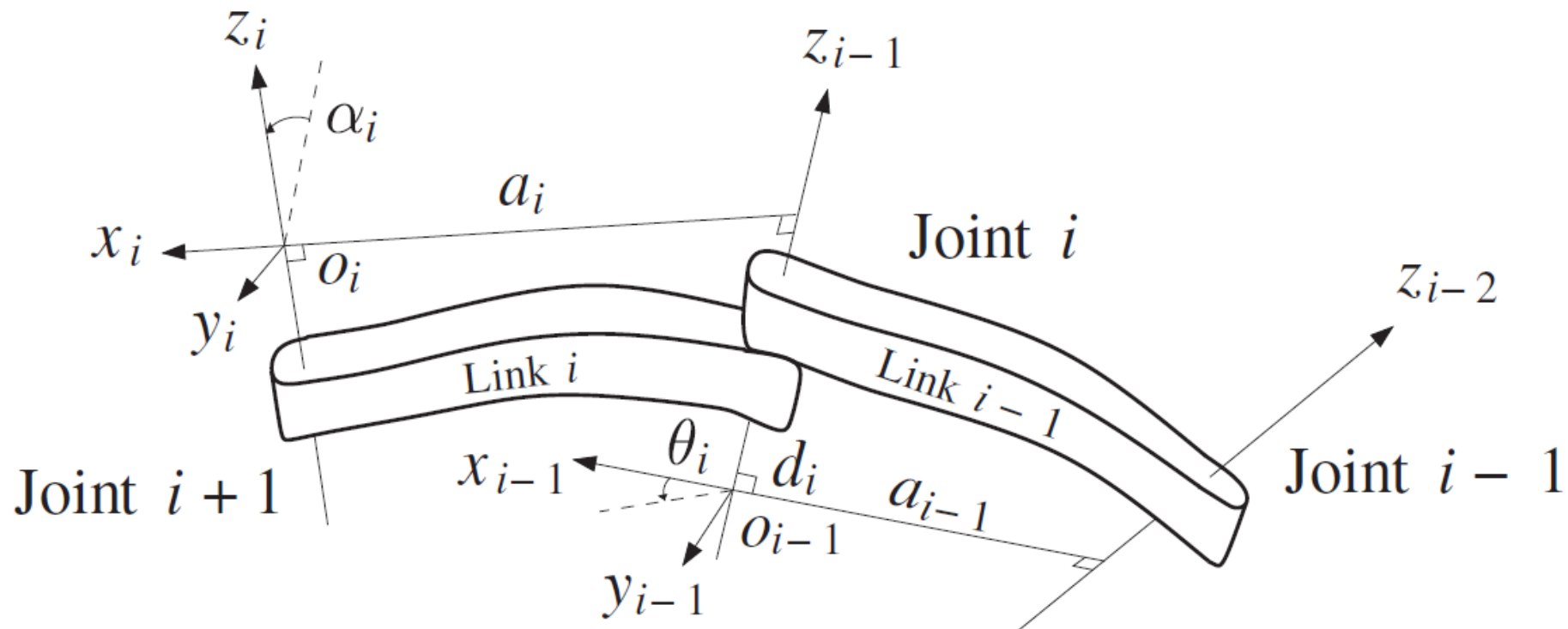
Assigning the coordinate frames

- We assign z_i to be the axis of actuation for joint $i + 1$.
- Thus, z_0 is the axis of actuation for joint 1, z_1 is the axis of actuation for joint 2, etc.
- There are two cases to consider:
 - (i) if joint $i + 1$ is revolute, z_i is the axis of revolution of joint $i + 1$;
 - (ii) if joint $i + 1$ is prismatic, z_i is the axis of translation of joint $i + 1$.
- Once we have established the z -axes for the links, we establish the base frame.
- We may choose the origin o_0 of the base frame to be any point on Z_0 .
- We then choose x_0, y_0 in any convenient manner so long as the resulting frame is right-handed.



Assigning the coordinate frames

- Once frame 0 has been established, we begin an iterative process in which we define frame i using frame $i - 1$, beginning with frame 1.





Assigning the coordinate frames

- To set up frame i it is necessary to consider three cases:
 - (i) the axes z_{i-1} , z_i are not coplanar,
 - (ii) the axes z_{i-1} , z_i intersect
 - (iii) the axes z_{i-1} , z_i are parallel.
- Note that in both cases (ii) and (iii) the axes z_{i-1} and z_i are coplanar.



Assigning the coordinate frames

(i) z_{i-1} and z_i are not coplanar:

- If z_{i-1} and z_i are not coplanar, then there exists a unique line segment perpendicular to both z_{i-1} and z_i such that it connects both lines and it has minimum length.
- The line containing this common normal to z_{i-1} and z_i defines x_i , and the point where this line intersects z_i is the origin o_i .
- By construction, both conditions (DH1) and (DH2) are satisfied and the vector from o_{i-1} to o_i is a linear combination of z_{i-1} and x_i .
- The specification of frame i is completed by choosing the axis y_i to form a right-handed frame.



Assigning the coordinate frames

(ii) z_{i-1} is parallel to z_i :

- If the axes z_{i-1} and z_i are parallel, then there are infinitely many common-normals between them and condition (DH1) does not specify x_i completely.
- In this case we are free to choose the origin o_i anywhere along z_i . One often chooses o_i to simplify the resulting equations.
- The axis x_i is then chosen either to be directed from o_i toward z_{i-1} , along the common normal, or as the opposite of this vector.
- A common method for choosing o_i is to choose the normal that passes through o_{i-1} as the x_i axis; o_i is then the point at which this normal intersects z_i . In this case, $d_i = 0$.
- Once x_i is fixed, y_i is determined, as usual by the right-hand rule.
- Since the axes z_{i-1} and z_i are parallel, α_i will be zero in this case.



Assigning the coordinate frames

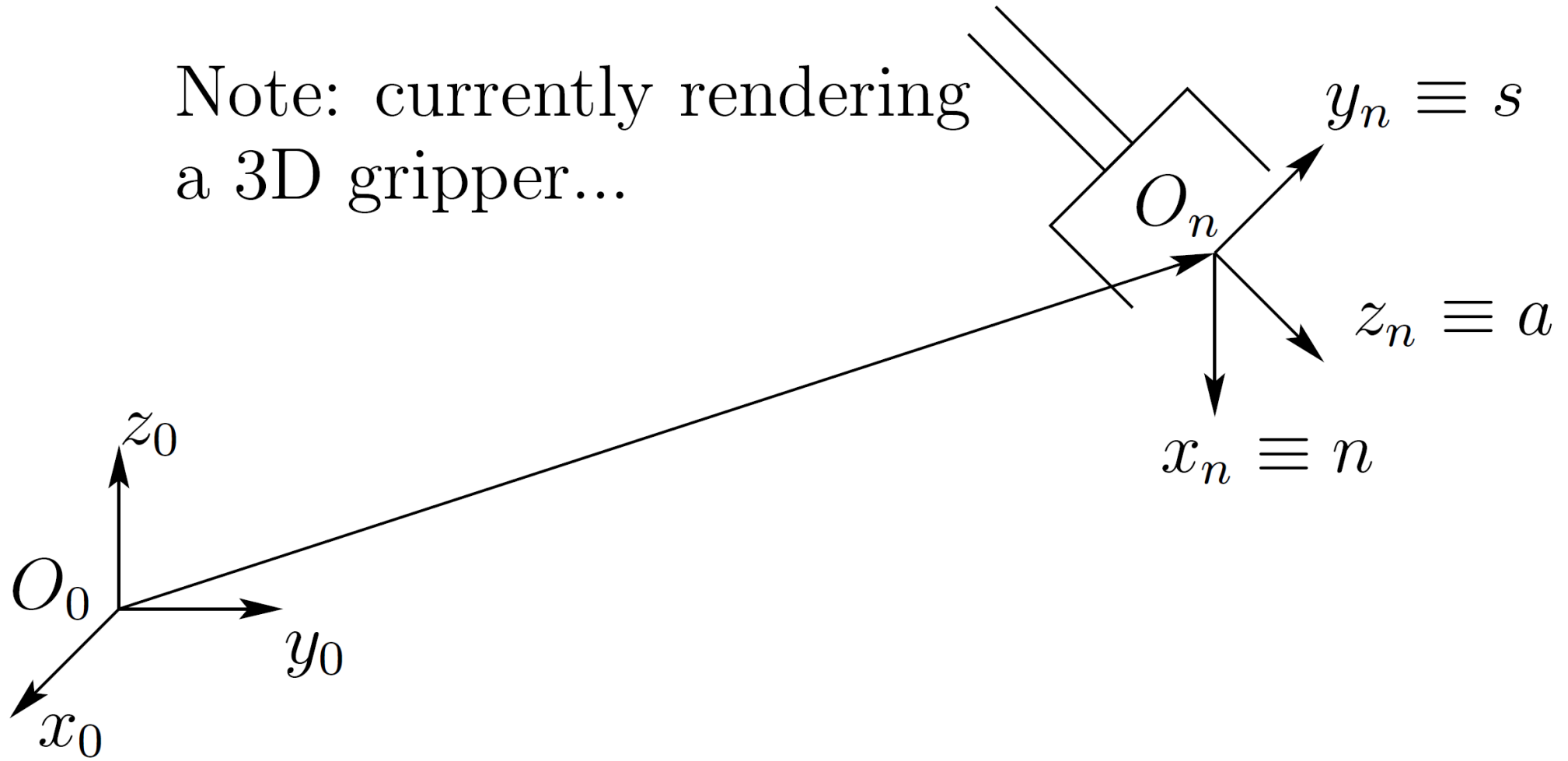
(iii) z_{i-1} intersects z_i :

- In this case x_i is chosen normal to the plane formed by z_i and z_{i-1} .
- The positive direction of x_i is arbitrary.
- The most natural choice for the origin o_i in this case is at the point of intersection of z_i and z_{i-1} .
- However, any convenient point along the axis z_i suffices.
- Note that in this case the parameter a_i equals 0.



Assigning the coordinate frames

Note: currently rendering
a 3D gripper...





Assigning the coordinate frames

- This constructive procedure works for frames $0, \dots, n - 1$ in an n -link robot.
- To complete the construction, it is necessary to specify frame n .
- The final coordinate system $o_n x_n y_n z_n$ is commonly referred to as the end-effector or tool frame.
- The origin o_n is most often placed symmetrically between the fingers of the gripper.
- The unit vectors along the x_n , y_n , and z_n axes are labeled as n , s , and a , respectively.
- The terminology arises from fact that the direction a is the approach direction, in the sense that the gripper typically approaches an object along the a direction.
- Similarly, the s direction is the sliding direction, the direction along which the fingers of the gripper slide to open and close, and n is the direction normal to the plane formed by a and s .

Summary of Forward Kinematic based on the DH convention



Step 1: Locate and label the joint axes $z_0, \dots, z_{n-1}$.

Step 2: Establish the base frame. Set the origin anywhere on the z_0 -axis. The x_0 and y_0 axes are chosen conveniently to form a right-handed frame.

For $i = 1, \dots, n - 1$ perform Steps 3 to 5.

Step 3: Locate the origin o_i where the common normal to z_i and z_{i-1} intersects z_i . If z_i intersects z_{i-1} locate o_i at this intersection. If z_i and z_{i-1} are parallel, locate o_i in any convenient position along z_i .

Step 4: Establish x_i along the common normal between z_{i-1} and z_i through o_i , or in the direction normal to the $z_{i-1} - z_i$ plane if z_{i-1} and z_i intersect.

Step 5: Establish y_i to complete a right-handed frame.

Summary of Forward Kinematic based on the DH convention



Step 6: Establish the end-effector frame $o_n x_n y_n z_n$. Assuming the n^{th} joint is revolute, set $z_n = a$ parallel to z_{n-1} . Establish the origin o_n conveniently along z_n , preferably at the center of the gripper or at the tip of any tool that the manipulator may be carrying. Set $y_n = s$ in the direction of the gripper closure and set $x_n = n$ as $s \times a$. If the tool is not a simple gripper set x_n and y_n conveniently to form a right-handed frame.

Summary of Forward Kinematic based on the DH convention



Step 7: Create a table of DH parameters a_i , d_i , α_i , θ_i .

a_i = distance along x_i from the intersection of the x_i and z_{i-1} axes to o_i .

d_i = distance along z_{i-1} from o_{i-1} to the intersection of the x_i and z_{i-1} axes. If joint i is prismatic, d_i is variable.

α_i = the angle from z_{i-1} to z_i measured about x_i .

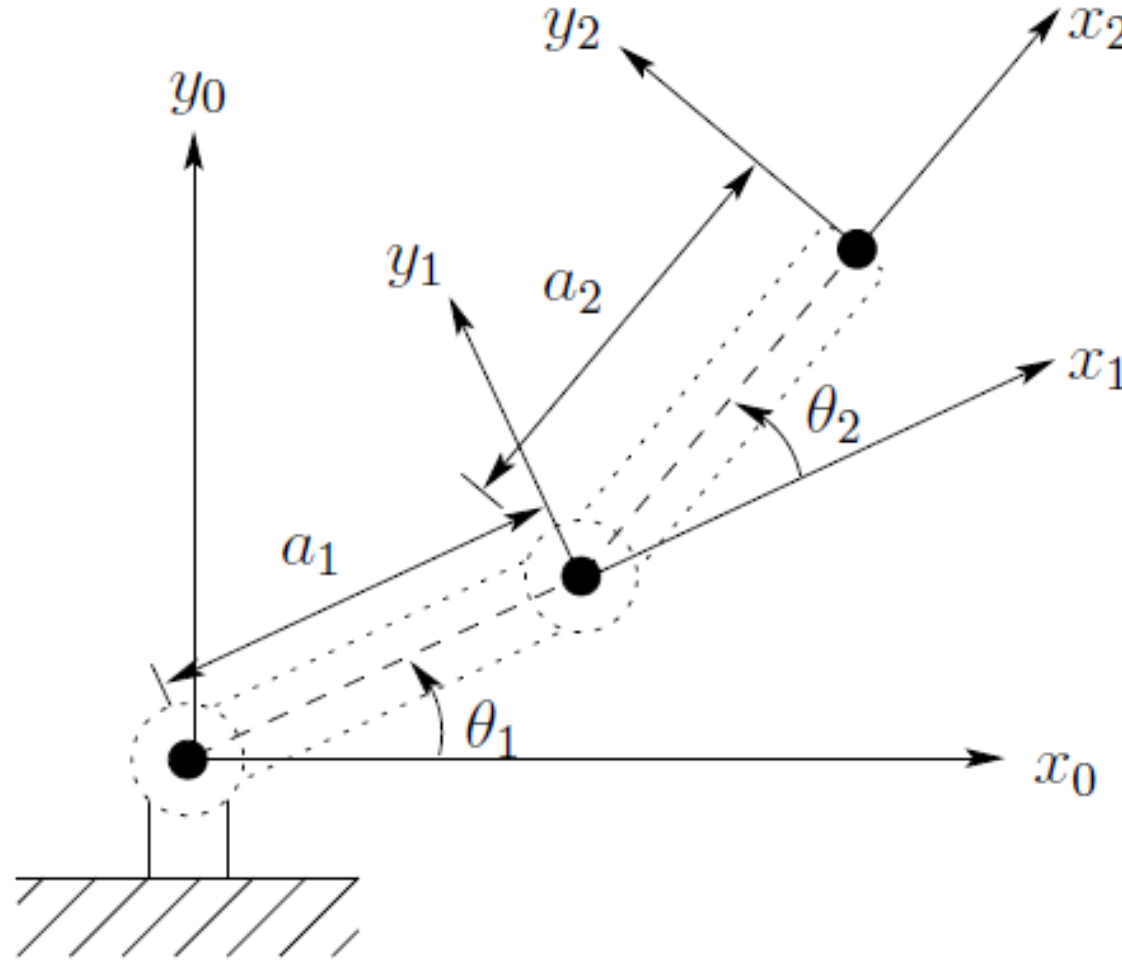
θ_i = the angle from x_{i-1} to x_i measured about z_{i-1} . If joint i is revolute, θ_i is variable.

Step 8: Form the homogeneous transformation matrices A_i by substituting the above parameters into Equation (3.10).

Step 9: Form $T_n^0 = A_1 \cdots A_n$. This then gives the position and orientation of the tool frame expressed in base coordinates.



Example 1 Planar Elbow Manipulator



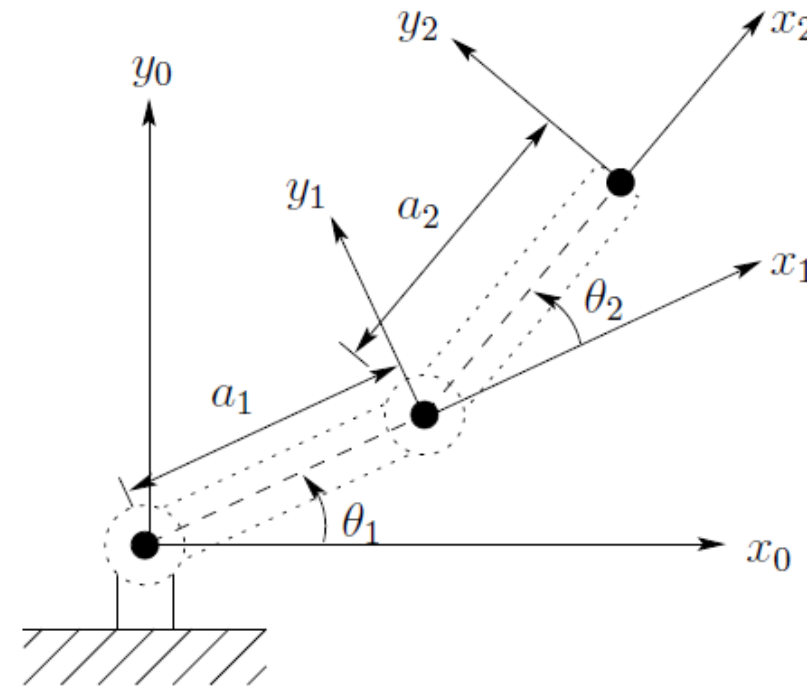


Example 1 Planar Elbow Manipulator

Link	a_i	α_i	d_i	θ_i
1	a_1	0	0	θ_1^*
2	a_2	0	0	θ_2^*

$$A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$A_2 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$





Example 1 Planar Elbow Manipulator

$$T_1^0 = A_1$$
$$T_2^0 = A_1 A_2 = \begin{bmatrix} c_{12} & -s_{12} & 0 & a_1 c_1 + a_2 c_{12} \\ s_{12} & c_{12} & 0 & a_1 s_1 + a_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

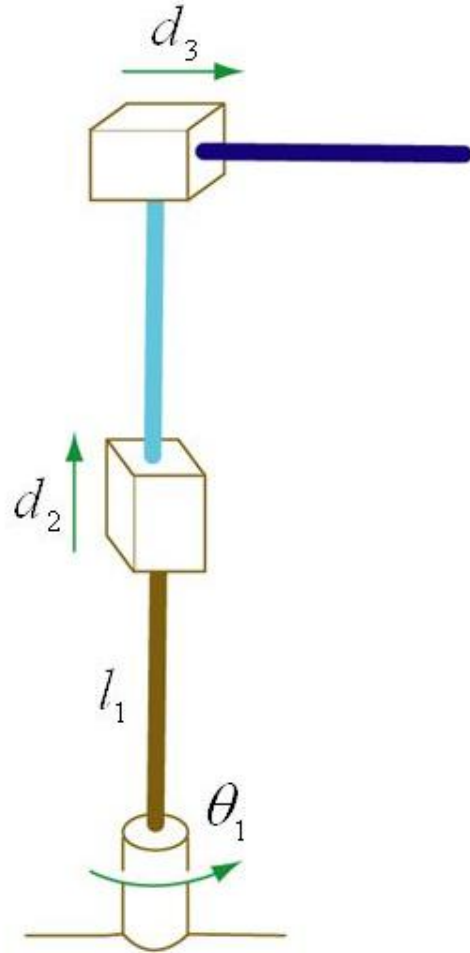
- Notice that the first two entries of the last column of T_2^0 are the x and y components of the origin o_2 in the base frame; that is,

$$x = a_1 c_1 + a_2 c_{12}$$

$$y = a_1 s_1 + a_2 s_{12}$$

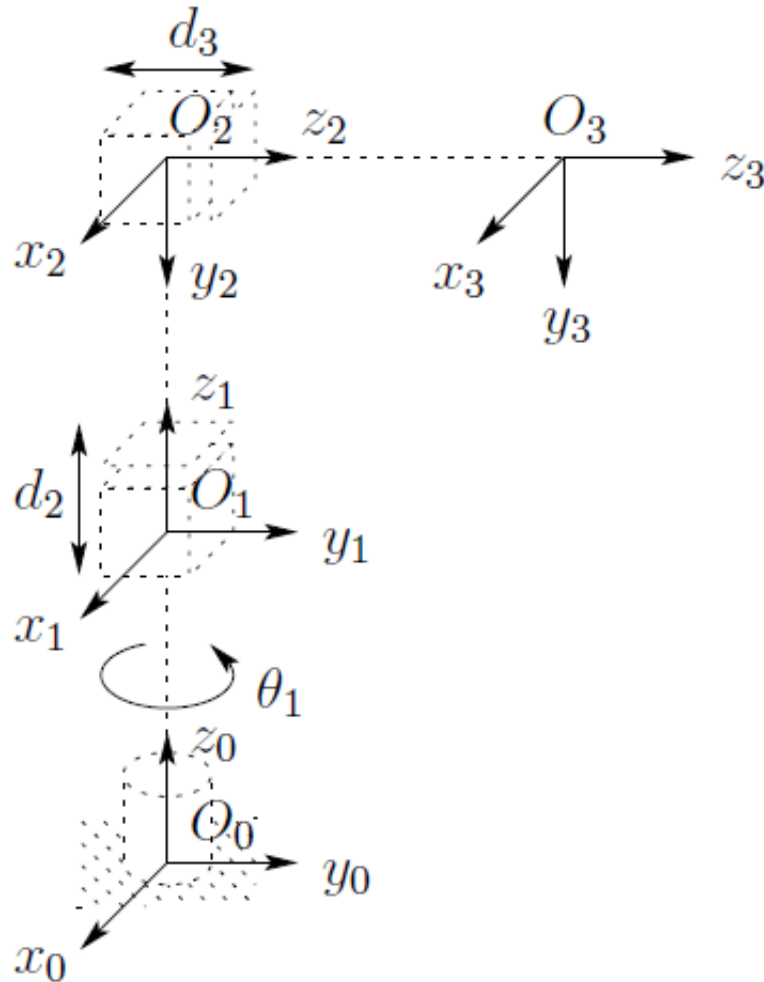


Example 2 Three-Link Cylindrical Robot





Example 2 Three-Link Cylindrical Robot



Link	a_i	α_i	d_i	θ_i
1	0	0	d_1	θ_1^*
2	0	-90	d_2^*	0
3	0	0	d_3^*	0

* variable



Example 2 Three-Link Cylindrical Robot

$$A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = A_1 A_2 A_3 = \begin{bmatrix} c_1 & 0 & -s_1 & -s_1 d_3 \\ s_1 & 0 & c_1 & c_1 d_3 \\ 0 & -1 & 0 & d_1 + d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link	a_i	α_i	d_i	θ_i
1	0	0	d_1	θ_1^*
2	0	-90	d_2^*	0
3	0	0	d_3^*	0

* variable



Any Questions